



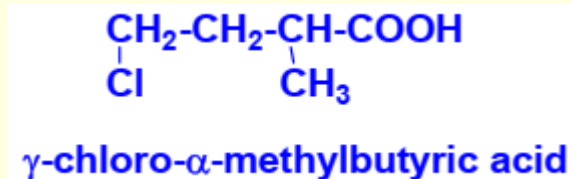
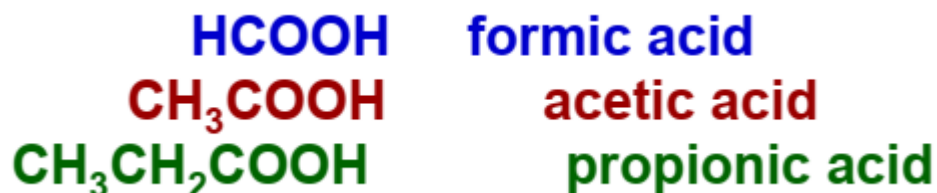
CARBOXYLIC ACID

RCOOH



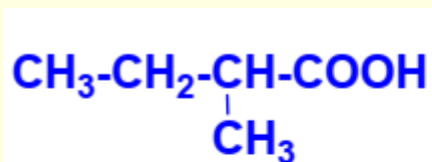
Danh pháp

Tên
thông
thường

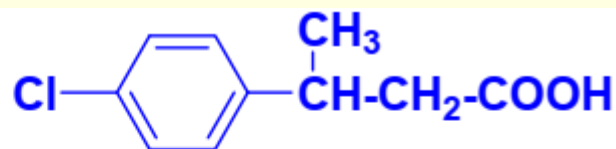


Tên
IUPAC

- Mạch chính dài nhất chứa nhóm -COOH (C1)
- Gọi theo tên hydrocarbon tương ứng, đổi ne $\rightarrow$ noic acid



2-methylbutanoic acid



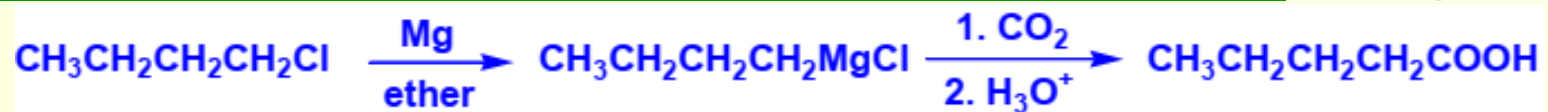
3-(p-chlorophenyl)butanoic acid



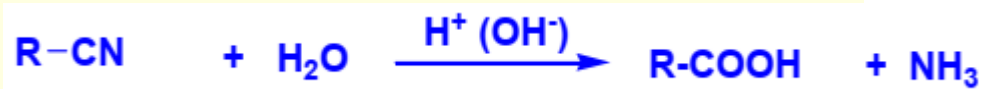
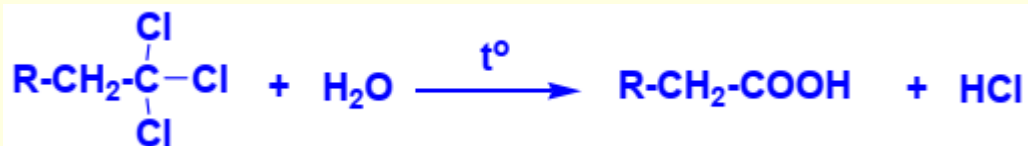
Điều chế



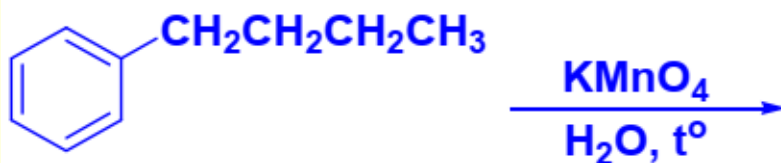
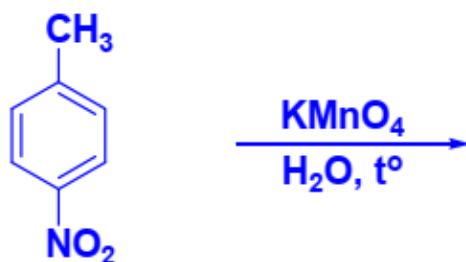
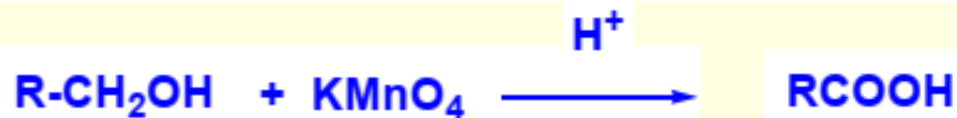
- Grignard



- Thủy phân



- Oxy hóa



Tính acid

Hút → tăng tính acid



Tính acid: $\text{F}_3\text{C-COOH} > \text{F}_3\text{C-CH}_2\text{-COOH} >$



Tính acid: $\text{CH}_3\text{-CH=CH-CH}_2\text{-COOH}$ (pKa 4.48) >
 $\text{CH}_2\text{=CH-CH}_2\text{-CH}_2\text{-COOH}$ (4.68) >
 $\text{CH}_3\text{-CH}_2\text{-CH=CH-COOH}$ (4.83)

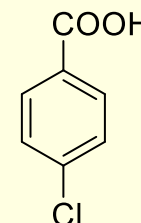
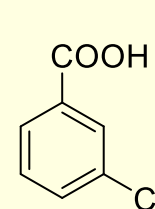
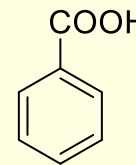
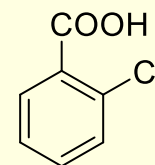
• Tính acid: $\text{CH}\equiv\text{C-COOH}$ (pKa 1.84) >
 $\text{CH}_3\text{-C}\equiv\text{C-COOH}$ (2.60) > $\text{CH}_2\text{=CH-COOH}$ (4.25)³⁴

* Acid có vòng thơm:

• Tính acid H-COOH (pKa 3.75) > $\text{C}_6\text{H}_5\text{-COOH}$ (4.18)
do +C của $\text{C}_6\text{H}_5\text{-}$ mạnh hơn -I

• Tính acid tùy thuộc bản chất & vị trí nhóm thế:
 $\text{o-NO}_2\text{-C}_6\text{H}_5\text{-COOH} > \text{p-} > \text{m-}$

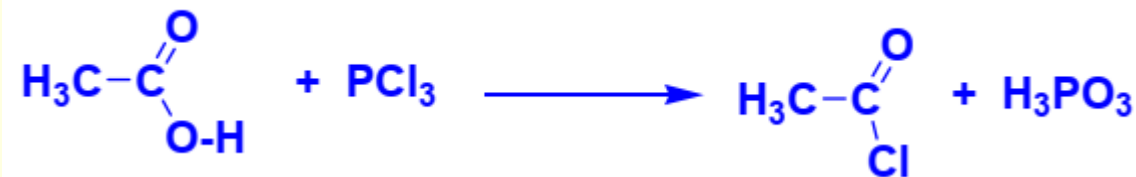
■ Ví dụ



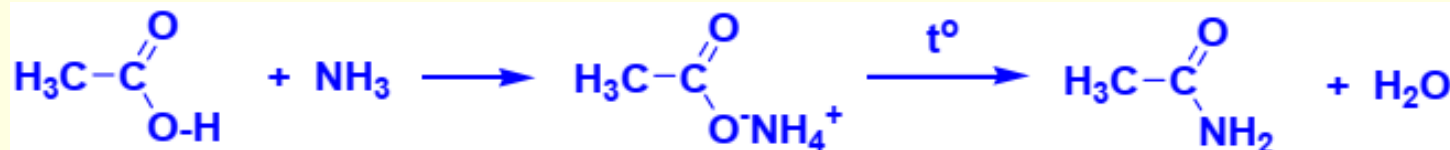
Tính chất hóa học



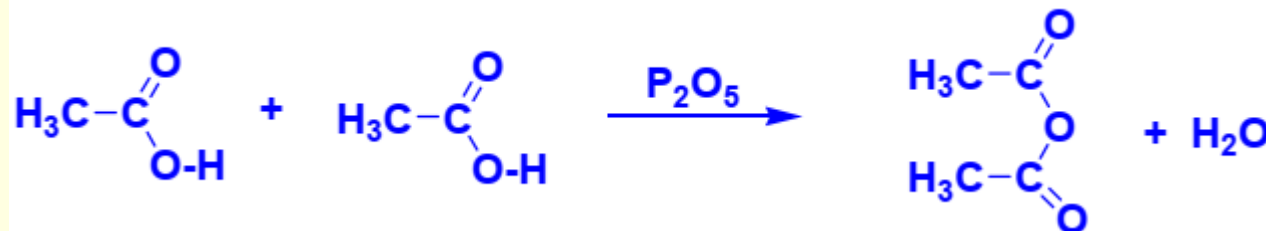
- Tạo acyl chloride



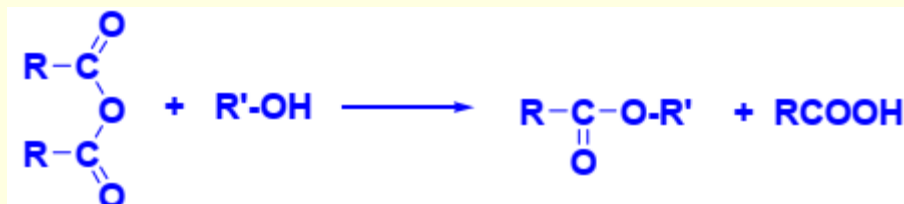
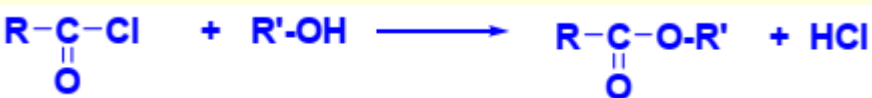
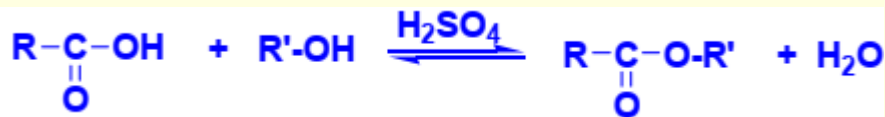
- Tạo amine



- Tạo anhydride



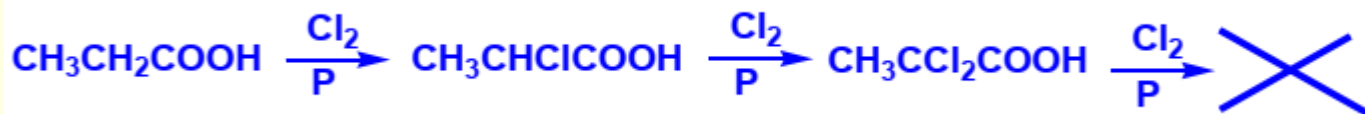
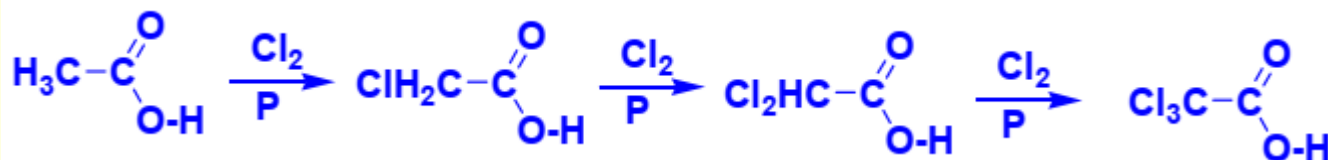
- Tạo ester



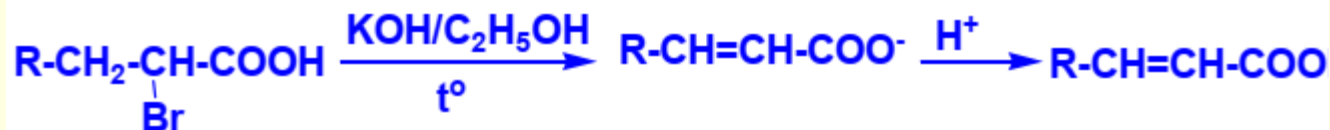
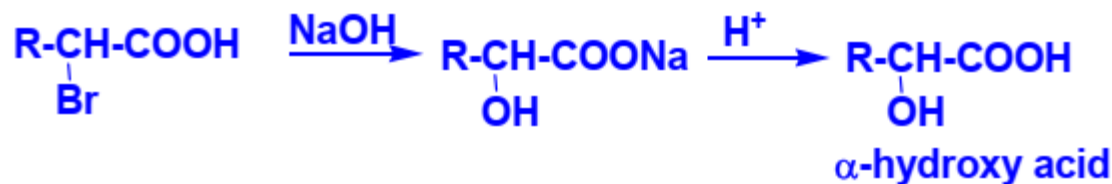
Tính chất hóa học



- Phản ứng thế H_α



- Tách và thế với dẫn xuất halogen tạo thành



- Khử tạo alcohol



Summary



- Điều chế:
 - $\text{RMgCl (+CO}_2\text{)}, \text{RCN thủy phân (H}_3\text{O}^+) \rightarrow \text{RCOOH}$
 - Oxy hóa (KMnO_4) $\text{ROH} \rightarrow \text{RCOOH}$, $\text{C}_6\text{H}_5\text{CHR}_2 \rightarrow \text{C}_6\text{H}_5\text{COOH}$ (phải còn H benzyl)
- Tính chất hóa học:
- Tính acid: Phản ứng với base: p/ứng với NaOH , NH_3 , ROH ($\text{RCOCl} > (\text{RCO})_2\text{O} > \text{RCOOH}$)
- Phản ứng thế H_α $\text{RCH}_2\text{COOH} + \text{Cl}_2/\text{Br}_2 (\text{P}) \rightarrow \text{RCHBrCOOH}$
- Phản ứng khử (LiAlH_4) $\rightarrow \text{ROH}$

Ví dụ

